Master's Thesis

Improving Odometry in Rimless Wheeled Rover using Sensor Fusion

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Abstract

Reliable state estimation is vital for Rimless Wheeled Rover (RWR) operating in unstructured planetary environments where external positioning systems such as Global Navigation Satellite System (GNSS) and motion capture systems are unavailable. One of the main challenges in dead-reckoning odometry approaches is the significant error accumulation due to sensor drift and unaccounted wheel slippage on loose terrains such as sand and gravel. This thesis presents an odometry approach using sensor fusion techniques within an Invariant Extended Kalman Filter (InEKF) framework. This approach is developed for the RWR platform, where the unique Rimless Wheel (RW) geometry and the associated wheel-ground contact dynamics make the existing baseline odometry susceptible to wheel slippage and drift. The proposed approach uses the Inertial Measurement Unit (IMU) and joint encoder measurements to construct predicted and measured Potential Contact Velocity (PCV) vectors, which are then used in the update step of the InEKF to estimate the rover state. The inconsistent joint encoder measurements compared to the IMU measurements are rejected before the update step of the InEKF. Which prevents the erroneous joint encoder measurements from being incorporated into the state estimation process. The pose estimates from the slip-aware odometry estimator are evaluated against the pose estimates from the motion capture system which is considered as the ground truth. The proposed approach is also compared with the baseline odometry currently integrated into the Coyote 3 RWR platform. The evaluation is performed through extensive real-world experiments with different trajectories on indoor floor and sand terrain. The evaluation results are presented as the Absolute Pose Error (APE), Relative Pose Error (RPE), drift rate and other statistical performance metrics.

The results show that... WIP...

Acknowledgements

Thanks to ....

Contents

List of Acronyms

APE Absolute Pose Error.

CVO Contact Velocity Odometry.

CVOE Contact Velocity Odometry Estimator.

DFKI Deutsches Forschungszentrum fur Künstliche Intelligent GmbH.

GNSS Global Navigation Satellite System.

GPS Global Positioning System.

IMU Inertial Measurement Unit.

InEKF Invariant Extended Kalman Filter.

KDL Kinematics and Dynamics Library.

LF Lowest Foot.

MD Mahalanobis Distance.

PC Potential Contact.

PCV Potential Contact Velocity.

RIC Robotics Innovation Center.

RPE Relative Pose Error.

RW Rimless Wheel.

RWR Rimless Wheeled Rover.

URDF Unified Robot Description Format.

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Conclusion and Future Work

This chapter summarizes the findings, observed improvements, and limitations of the proposed approach as well as directions for future work. This chapter is organized into three sections: Conclusion, Limitations, and Future Work. The Conclusion section details the improvements and drawbacks of the Contact Velocity Odometry (CVO) approach observed during experimental evaluation. This section also summarizes the problem, the methodology used to address it, and the results obtained. The Limitations section describes the limitations of the proposed approach. The limitations of the evaluation methods and specific assumptions used in this work are also discussed. The Future Work section presents potential approaches for addressing these limitations and identifies further research directions.

1.1 Conclusion

The current baseline odometry uses a circular-wheel approximation and does not explicitly account for wheel slip. Hence, this thesis investigated an Invariant Extended Kalman Filter (InEKF)-based sensor fusion approach that accounts for the contact-point velocity of the rimless-wheel spokes during state estimation. The existing InEKF library was extended to implement the proposed Potential Contact Velocity (PCV)-based measurement update.

In the proposed approach, the Inertial Measurement Unit (IMU) measurements are used to propagate the filter state. The predicted PCV is calculated using the propagated state and the spoke-tip positions, while the measured PCV is calculated using the wheel joint encoder measurements and forward kinematics. For each selected contact spoke, the squared Mahalanobis Distance (MD) is calculated using the residual and its innovation covariance. Measurements below the specified threshold are used in the left-invariant correction step, while measurements above the threshold are rejected as inconsistent. The proposed approach was evaluated using datasets collected on flat-floor, sand, and crater terrains. Metrics such as travel-distance error, final-position error, yaw Root Mean Square Error (RMSE), z-position RMSE, and translational and rotational Relative Pose Error (RPE) drift RMSE were used to quantify its performance. The developed framework also includes Lowest Foot (LF)- and MD-based contact-spoke selection and a configurable setup for rejecting inconsistent PCV measurements before the filter correction.

The evaluation results show that the proposed CVO approach does not consistently outperform the baseline odometry. On flat-floor terrain, the baseline odometry generally provides lower translational and rotational RPE drift, indicating that the PCV-based update does not provide a consistent advantage under the selected flat-floor conditions. On sand terrain, the CVO approach provides lower translational RPE drift in six of the seven selected datasets. However, its rotational RPE drift is higher in six of the seven datasets. These results indicate that the proposed approach is more beneficial for local translational motion estimation on deformable terrain than for local rotational estimation.

The crater terrain experiments demonstrate the potential benefit of rejecting inconsistent PCV measurements under slip-prone conditions. In `DATA_26_01`, enabling slip elimination substantially reduces both the translational and rotational RPE drift compared with the same CVO configuration without slip elimination. In `DATA_26_04`, enabling slip elimination produces little change in the translational RPE drift and slightly increases the rotational RPE drift. Compared with the baseline odometry, the CVO approach with slip elimination provides lower translational RPE drift in both selected crater datasets but higher rotational RPE drift. Therefore, the effectiveness of measurement rejection depends on the characteristics of the inconsistent wheel measurements and the motion condition.

Due to time constrains, the rotation in-place, move forward when pulled back in both flat floor terrain and sand terrain datasets were not evaluated. The evaluation of these datasets could provide further insight into the performance of the proposed approach under different motion conditions. The metrics and plots to compare the contact-spoke selection methods such as LF and MD are developed and tested during the implementation of the proposed approach. However, due to the lack of ground truth information about the contact estimation, the accuracy of the contact-spoke selection methods is not presented in this work. A sample contact estimation comparison plot between the LF and MD methods and the corresponding accuracy metrics plot are presented in Appendix **??**.

Overall, this thesis demonstrates that PCV measurements and innovation-based measurement rejection can improve local translational odometry on deformable terrain and under conditions with pronounced slip. However, the proposed approach does not yet provide consistent improvement across all terrains and motion types, and rotational estimation remains an important weakness. The work developed an odometry framework for the Coyote 3 rover that is contact-aware and rejects wheel slip, while achieving a consistently better replacement for the baseline odometry remains future work.

1.2 Limitations

- The proposed approach uses the squared MD to detect inconsistent PCV measurements. However, a high MD value can be caused by wheel slip, incorrect contact-spoke selection, modelling error, sensor noise, transformation error, or incorrect covariance values. Therefore, the rejected measurements cannot always be confirmed as physical wheel slip.
- In addition, the Vicon system provides the ground truth pose of the rover body, but it does not provide ground truth information about wheel slip or contact information. Hence, the accuracy of slip detection and contact-spoke estimation cannot be calculated directly.
- The LF-based contact estimation also assumes that the vertically lowest spoke is in contact, which may not be correct on inclined, rough, deformable, or obstacle-covered terrain. Similarly, the MD-based method can select a spoke with a low residual, but this does not guarantee that the selected spoke is physically in contact.

- Using the MD both to select the contact spoke and to determine the measurements used in the PCV update creates a coupled dependency between contact estimation and the measurement update. During testing, this configuration resulted in poor performance of the CVO approach. Hence, the MD-based contact estimation was disabled in the evaluated implementation, and only the LF-based contact estimation was used.

- The chi-squared threshold and measurement noise values used in the evaluated implementation are fixed and were selected experimentally. The contact MD threshold used during the testing of MD-based contact estimation was also selected experimentally. Their performance was not systematically evaluated for every terrain type, velocity, motion type, and contact condition. The current approach also performs complete measurement rejection, where a measurement is either used or rejected. The magnitude and direction of slip are not estimated.

- If several wheel measurements are rejected at the same time, the filter depends mainly on the IMU propagation step, which can increase position and orientation drift. The evaluation results also show that the rotational RPE drift is higher for the CVO approach in most of the selected flat-floor and sand terrain datasets. This shows that orientation estimation, specifically yaw estimation remains a weakness of the proposed approach.

- The effect of slip elimination was explicitly compared only in the crater terrain experiments. Only four crater datasets were collected, and two datasets were discussed in detail. The improvement was substantial in `DATA_26_01`, but small in `DATA_26_04`. Therefore, the crater results are not sufficient to make a general conclusion about all inclined and slip-prone terrain conditions.

- The effect of different Rimless Wheel (RW) gait configurations were not evaluated separately. Different gaits can change the spoke contact timing, the number of simultaneous contacts, and the calculated PCV measurements. Changes in gait can also introduce different slip characteristics, which can affect the performance of the proposed approach. These effects were not evaluated in this work.

- Although three repetitions were collected for most dataset types, the evaluation does not present aggregate statistics across all repetitions. Therefore, the reported results mainly describe the selected datasets and are not sufficient to establish statistically repeatable improvements.

1.3 Future Work

From the collected datasets, the visually identifiable slip events and contact-spoke information could be extracted from the video recordings collected during the experiments. This information could be used to evaluate the accuracy of the contact-spoke estimation methods and the slip detection method. However, due to the lack of time, this was not performed in this work. Improving the dataset with this information from video analysis would allow the proposed approach to be evaluated more thoroughly and also help improve the CVO approach.

The multi-spoke contact estimation method is developed in this work, but it was not evaluated due to time constraints. The multi-spoke contact estimation method could be evaluated using the collected datasets. A detailed evaluation comparing the single and multi-spoke contact estimation methods could be performed to determine the effect of multi-spoke contact estimation on the CVO performance. To avoid jittery contact-spoke selection, the previous contact-spoke selections could be stored and based on the wheel rotation direction, the contact-spoke selection could be limited to the adjacent spokes of the previously selected spoke.

The contact estimation could also be changed from binary contact selection to a probabilistic contact estimation method. A probabilistic contact estimation method could assign a contact probability to each spoke and use this information in the PCV update. Currently, the measurement uncertainty is fixed at the configuration time and never changes during the filter operation. A dynamic measurement uncertainty could be used to inform the filter about the confidence of the measurement when squared MD values are high but not above the rejection threshold.

The noise parameters used in the PCV update could be tuned to improve the performance of the proposed approach. Currently, the noise values are selected experimentally and kept fixed during the filter operation. The noise values could be tuned based on the terrain type and could be switched based on the terrain type during operation. The effects of different RW gait configurations could be evaluated by repeating the same trajectories with different predefined wheel positions. This would allow the effects of gait on contact timing, simultaneous contacts, wheel slip, and PCV consistency to be studied.

A slip magnitude and direction estimation method in addition to slip rejection could be developed to improve the performance of the proposed approach. This information could be used to distinguish longitudinal slip, which occurs along the wheel rolling direction, from lateral slip, which occurs perpendicular to the wheel rolling direction. Lateral slip can be more noticeable during turning, while longitudinal slip commonly occurs during forward or backward traction. Estimating the slip direction could allow direction-dependent measurement uncertainty to be used and may improve yaw estimation during turning motions.